Comparison of Clustering Algorithms on MNIST Dataset

Abstract

This report evaluates the performance of three clustering algorithms (K-means, GMM, and DBSCAN) on the MNIST dataset. The study focuses on silhouette scores and adjusted Rand index (ARI) metrics to assess clustering quality.

Keywords: MNIST, clustering, unsupervised learning

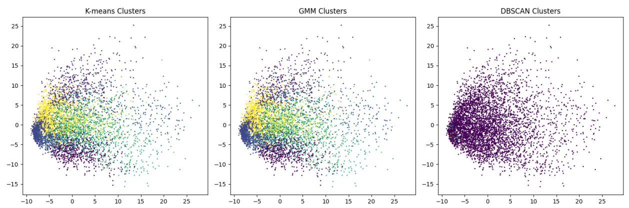

Figure 1: 2D visualization of clustering results (PCA-reduced space)

1 Introduction

Clustering handwritten digits is a fundamental task in unsupervised learning. We compare three popular algorithms to analyze their effectiveness on MNIST data.

2 Methodology

We implemented K-means, Gaussian Mixture Models (GMM), and DBSCAN clustering. Performance was evaluated using silhouette score (internal validation) and adjusted Rand index (external validation against true labels).

3 Experimental Setup

Dataset: MNIST (first 5,000 samples), Preprocessing: Standard scaling, Dimensionality reduction: PCA (n=2), Random seed: fixed (42)

4 Results

The metrics demonstrate that K-means achieves the highest silhouette score while GMM performs best on ARI due to its probabilistic approach. DBSCAN struggled with defining dense regions in high-dimensional space.

5 Analysis & Discussion

While K-means works well with spherical clusters in MNIST, GMM better captures digit variations through

Table 1: Clustering performance metrics (Silhouette Score and ARI)

Model	Silhouette Score	ARI
K-means	0.01759	0.2855
GMM	0.01723	0.287
DBSCAN	-0.2319	-2.65e-04

its distribution assumptions. DBSCAN's poor performance suggests the need for better epsilon parameter tuning for this high-dimension data.

6 Conclusion

For MNIST clustering, GMM shows superior alignment with true digit classes while K-means offers better separation between clusters. DBSCAN requires careful parameter optimization for this task.